Surface-Object Detection and Tracking from Commercial X-Band Radar Using Adaptive ST-DBSCAN

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Motivation

Commercial X-Band Radar Challenges:

- No Doppler phase (intensity only)
- Non-stationary sea clutter
- Heavy-tailed amplitude statistics
- Range-varying point density (R^{-4})
- Intermittent target visibility

DBSCAN Limitations:

- Fixed ε fails with heterogeneous density
- Track fragmentation from dropouts
- No temporal linkage

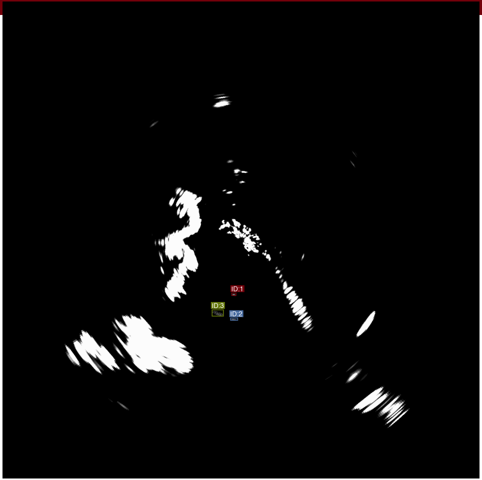


Figure: Synthetic radar frame

Related Work

Classical Tracking (1960s-1980s):

- Kalman filtering (1960)
- MHT - Reid (1979)
- JPDAF - Fortmann (1983)
- CFAR detection - Rohling (1983)
- Issue: Assumes known clutter distributions

Density-Based Clustering (1990s-2000s):

- DBSCAN - Ester et al. (1996)
- OPTICS - Ankerst et al. (1999)
- VDBSCAN - Liu et al. (2007)
- ST-DBSCAN - Birant & Kut (2007)
- HDBSCAN - Campello et al. (2013)

Modern Deep Learning (2010s):

- SORT - Bewley et al. (2016)
- DeepSORT - Wojke et al. (2017)
- Issue: Require appearance features (not available in radar)

Our Contribution:

- Adaptive ε via k-NN statistics
- Gap recovery for track continuity
- Real-time embedded implementation
- Union-Find structures for efficient merging

Problem Formulation

Detection Set at Sweep t:

$$\mathcal{D}_t = \{(r_i, \theta_i, I_i, t) : i = 1, \ldots, N_t\}$$

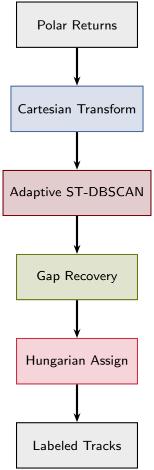
Objective: Partition into tracks:

$$\mathcal{D} = \mathcal{T}_1 \cup \mathcal{T}_2 \cup \cdots \cup \mathcal{T}_K \cup \mathcal{N}$$

Distance Metrics:

$$d_s(\mathbf{p}_i, \mathbf{p}_j) = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$$
$$d_t(\mathbf{p}_i, \mathbf{p}_j) = |t_i - t_j|$$

Processing Pipeline:



Adaptive ST-DBSCAN Method

Challenge: Fixed ε fails with density variation
Solution: k-NN adaptive epsilon

$$\varepsilon_s(\mathbf{p}) = \min\left(d_k(\mathbf{p}) \cdot \alpha, \varepsilon_{s,\max}\right)$$

Global estimation:

$$\varepsilon_s^{\text{global}} = \text{Percentile}_{90}\left(\{d_k(\mathbf{p}) : \mathbf{p} \in \mathcal{S}\}\right)$$

Epsilon floor:

$$\varepsilon_s^{\text{final}} = \max\left(\varepsilon_s^{\text{global}}, \varepsilon_{\min}\right)$$

Algorithm 1 *

Algorithm 1: Adaptive Epsilon

Require: $\mathcal{P}$, k, β, ρ, $\varepsilon_{\min}$, $\varepsilon_{\max}$
Ensure: ε_s
1: $\mathcal{S} \leftarrow$ sample $\lfloor \beta \cdot |\mathcal{P}| \rfloor$ points
2: **for all** $\mathbf{p} \in \mathcal{S}$ **do**
3: $d_k(\mathbf{p}) \leftarrow k$-th nearest neighbor distance
4: **end for**
5: $\varepsilon_s \leftarrow \text{Percentile}_\rho\left(\{d_k(\mathbf{p})\}\right)$
6: **return** $\max(\varepsilon_{\min}, \min(\varepsilon_s, \varepsilon_{\max}))$

Parameters: $k = 5$, $\alpha = 1.5$, $\rho = 90$, $\varepsilon_{\min} = 20$ px

Gap Recovery Mechanism

Problem: Intermittent visibility fragments tracks

Solution: Union-Find merging across temporal gaps

Temporal gap:

$$\Delta t = t_j^{\text{start}} - t_i^{\text{end}}$$

Spatial gap between centroids:

$$\Delta s = \|\bar{\mathbf{p}}_i(t_i^{\text{end}}) - \bar{\mathbf{p}}_j(t_j^{\text{start}})\|_2$$

Velocity-based threshold:

$$d_{\max} = v_{\max} \cdot \Delta t \cdot T_s$$

Merge if: $\Delta t \leq \tau$ and $\Delta s \leq d_{\max}$

Algorithm 2 *

Algorithm 2: Gap Recovery

Require: Clusters $\{\mathcal{T}_1, \ldots, \mathcal{T}_K\}$, τ, $v_{\max}$, T_s
Ensure: Merged $\{\mathcal{T}'_1, \ldots, \mathcal{T}'_{K'}\}$

Sort clusters by t^{end} ascending
Initialize union-find structure
for all cluster $\mathcal{T}_i$ **do**
    for all cluster $\mathcal{T}_j$ with $t_j^{\text{start}} > t_i^{\text{end}}$ **do**
        $\Delta t \leftarrow t_j^{\text{start}} - t_i^{\text{end}}$
        if $\Delta t > \tau$ **then**
            break
        end if
        $d_{\max} \leftarrow v_{\max} \cdot \Delta t \cdot T_s$
        $\Delta s \leftarrow \|\bar{\mathbf{p}}_i - \bar{\mathbf{p}}_j\|_2$
        if $\Delta s \leq d_{\max}$ **then**
            $\textsc{Union}(\mathcal{T}_i, \mathcal{T}_j)$
        end if
    end for
end for
return clusters grouped by components

Hungarian Assignment

Purpose: Frame-to-frame track association

Cost matrix between track centroids:

$$C_{ij} = \|\mathbf{c}_i^{t-1} - \mathbf{d}_j^t\|_2$$

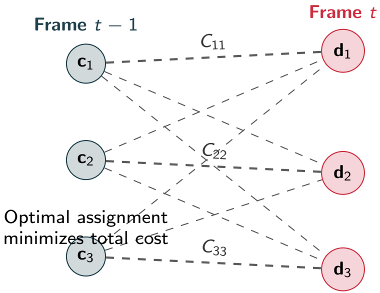
Optimal assignment:

$$\pi^* = \arg\min_{\pi} \sum_i C_{i,\pi(i)}$$

Gating: Reject assignments with $C_{ij} > \gamma$

Benefits:

- Low latency for real-time output
- Explicit track identity maintenance
- Complements ST-DBSCAN clustering



Implementation

Platform: NVIDIA Jetson AGX Orin

- Arm Cortex-A78AE (12 cores)
- 32 GB unified memory
- ˜25 W power consumption

Implementation: Rust

- Memory safety
- Zero-cost abstractions
- Predictable performance
- Spatial hashing ($h = 2\varepsilon_{s,\max}$)
- Rayon parallelism (k-NN queries)
- Union-Find (gap recovery)

Performance:

- **511 $\pm$ 93 ms per sweep**
- ˜2 sweeps/s
- Exceeds real-time for 48 RPM radar
- Memory $<$ 500 MB

Table: Per-Sweep Timing Breakdown

Stage	Time (ms)
Frame loading	7
Adaptive ε (k-NN)	196
ST-DBSCAN clustering	169
Cluster extraction	34
Gap recovery	19
Hungarian assignment	3
I/O (output)	40
Total	**511**

Bottlenecks:

- k-NN computation (38%)
- ST-DBSCAN clustering (33%)
- Both parallelized with Rayon

Clustering Comparison

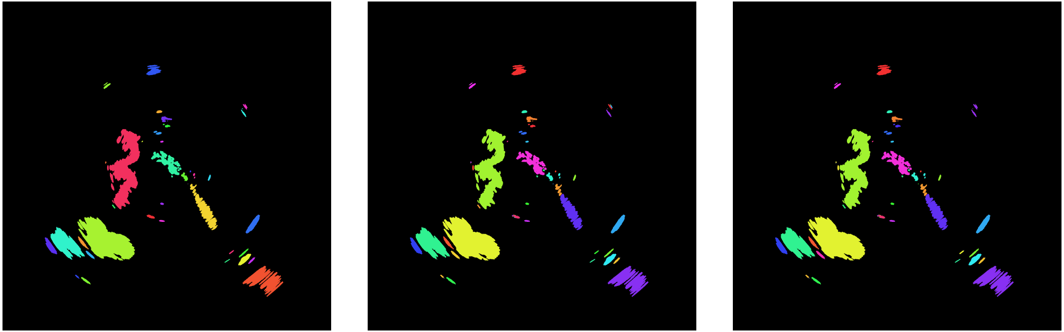


Figure: *

Left: Vanilla DBSCAN ($\varepsilon_s = 10$): 1693 clusters, many spurious detections. **Center:** Adaptive ST-DBSCAN ($\varepsilon_{\text{min}} = 20$): 1177 clusters, larger spatial neighborhoods. **Right:** Adaptive + Gap Recovery: merged temporally adjacent clusters.

Tracking Comparison

Figure: *

Left: Vanilla DBSCAN: 48 tracks, extensive false detections from land/clutter. **Center:** Adaptive ST-DBSCAN: 31 tracks, maintained target coverage. **Right:** Adaptive + Gap Recovery: 31 tracks, consolidated associations, zero ID switches.

Results: MOT Metrics

Table: Multi-Object Tracking Metrics on Synthetic Data

Method	MOTA	MOTP (px)	Recall	Prec.	FP	IDSW
Vanilla DBSCAN $\varepsilon_s = 10$	-6.97	1.80	0.987	0.112	1208	18
Standard ST-DBSCAN $\varepsilon_s = 10$, $\varepsilon_t = 2$	-0.10	–	0.000	0.000	16	0
Adaptive ST-DBSCAN $\varepsilon_{\min} = 20$	-4.44	**1.19**	**0.974**	0.155	816	18
Adaptive + Gap $\tau = 1$, $d = 0.5$	-3.82	**1.18**	0.812	**0.159**	**695**	**0**

Dataset:

- 50 frames
- 4 objects
- 15-40% dropout rate
- 1735x1735 px
- 1847 signatures

Key Findings:

- Negative MOTA: High-clutter environment
- Gap recovery eliminates ID switches
- 32% FP reduction over vanilla DBSCAN
- Best MOTP: 1.18 px localization

Results: Per-Object Performance

Table: Per-Object Tracking (Vanilla DBSCAN)

Object	Visible	Lifetime	MOTP (px)
Object 0	50	100%	1.01
Object 1	50	100%	1.41
Object 2	50	100%	1.16
Object 3	4	50%	47.2
Mean	–	87.5%	12.7

Table: Timing Summary

Metric	Value
Total per sweep	511 ± 93 ms
Throughput	~2 sweeps/s
Real-time req.	1.25 s/sweep
Memory usage	< 500 MB
Margin	**2.4**$\times$

Observations:

- 100% lifetime for persistent objects
- Sub-1.5 px MOTP for tracked objects
- Object 3: brief appearance (4 frames)
- Initialization latency affects short tracks

Deployment Viability:

- Exceeds real-time by 2.4$\times$
- Headroom for latency spikes
- Low memory footprint
- Suitable for shipboard deployment

Limitations and Future Work

Known Limitations:

- Epsilon floor required for dense data (less adaptive)
- Synthetic evaluation only (real radar pending)
- Target crossing/merging causes ID confusion
- Stationary objects hard to distinguish from clutter
- Hyperparameters (k, percentile) still manual
- Dense point clouds ($>$1M points) remain challenging

Failure Cases:

- Multiple targets within ε_s merge
- Extended path crossings cause persistent ID confusion
- Greedy Hungarian cannot disambiguate complex scenarios

Future Work:

- Kalman filtering for velocity estimation
- Multi-hypothesis tracking (MHT) for ambiguous associations
- AIS fusion for supervised track identification
- 3D extension for vertically-scanning radars
- Geographic masking to suppress land returns
- Real radar validation with ground truth
- Adaptive temporal threshold ε_t

Application Domains:

- Autonomous maritime navigation
- Collision avoidance systems
- Harbor surveillance
- Search and rescue

Questions?

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